

3.4.5 CNS 2205: Network Programming

Lecture Hours: 45

Prerequisites:

- CNS 1102: Introduction to programming

Purpose of the course

To provide students with an understanding of how the Unix operating system is deployed in data networking scenarios, and the ability to successfully deploy, program and manage such systems.

Expected Learning outcomes of the course:

At the end of this course, the student will be able to:

1. Demonstrate the use of network administration tools on Unix systems,
2. Understand the configuration of common network services, devices and security,
3. Design and program unfamiliar network services.

Course Content

The Unix CLI and shell: The Bourne Again Shell (BASH), Linux systems administration, Useful commands (awk, sed, rsync, apt-get, grep, date, man), Linux text editors (vi, vim, nano), Redirecting output, Task automation with cron; The Linux Kernel: What is the Kernel, User Space Vs Kernel Space, System call interface, Kernel Space, hardware, Linux Kernel Startup, Linux Kernel configuration, Make utilities, Building Kernel Modules, Writing Pseudo Drivers, Roles of Drivers, Building communication with Driver Modules; Daemons: Creating a daemon, SIGHUP; Unix/Linux network programming: Linux sockets programming, Sockets system calls, client/server, peer-to-peer, TCP sockets, UDP sockets, raw sockets, Populating protocol headers, TCP client, UDP client, TCP server, UDP server, Iterative server – TCP Iterative server, Concurrent server – TCP concurrent server; Multithreading and exception handling; Finger, DNS, HTTP, and ping clients and servers; Network traffic analysis: Linux packet capture library, tcpdump, wireshark, Linux traffic control; Configuring Unix networked services: LDAP, DNS, File and Print Sharing, Version control with SVN and Git,

SMTP, HTTP, NTP, Linux routers, Linux firewalls, Linux NAT/NAPT gateways; Monitoring and troubleshooting: ifconfig, ping, hping3, traceroute, tcptraceroute, nslookup, nmap, iperf, netstat, netcat, top, tcpdump, iptraf, nethogs, iftop, monitorix, arpwatch, suricata, nmon, collect, SNMP with Nagios and Cacti, vnStat php, ELK stack, System logs, syslog, Simple Object Access Protocol (SOAP), Java Server Pages (JSP), Java Server Faces (JSF).

Mode of Delivery

- Lectures and Tutorials and Practical Lab Sessions

Instructional Materials and/or equipment.

- Computers
- Network hardware

Course Assessment

- | | |
|-------------------------|-------------|
| • Examination | 60% |
| • Continuous Assessment | 40% |
| • Total | 100% |

Core Reading Materials

1. Advanced Programming in the UNIX Environment, 3rd ed., W. Richard Stevens, Stephen A. Raquo, Addison Wesley, 2013; ISBN-10: 0321637739
2. UNIX Network Programming, Volume 1, 3rd ed., W. Richard Stevens, Bill Fenner, and Andrew Rudoff, Addison Wesley, 2004.

Recommended Reference Material

1. Computer Networking: A Top-Down Approach (7th Edition), James Kurose and Keith Ross, Pearson, 7th ed. (2016), ISBN-10: 0133594149 (ISBN-13: 978-0133594140)
2. UNIX and Linux System Administration Handbook, Evi Nemeth et al., Addison-Wesley Professional; 5 ed., 2017, ISBN-10: 0134277554 (ISBN-13: 978-0134277554)

3. Linux: Beginners guide for learning Linux & Shell scripting, Scott Harvey, CreateSpace Independent Publishing Platform, 2017, ISBN-10: 1975877063 (ISBN-13: 978-1975877064)
4. Learning Linux Shell Scripting, Ganesh Sanjiv Naik, Packt Publishing, 2015, ISBN-10: 1785286218 (ISBN-13: 978-1785286216)
5. Linux Administration: A Beginners Guide, Wale Soyinka, McGraw-Hill Education; 6 ed., 2012, ISBN-10: 0071767584 (ISBN-13: 978-0071767583)
6. UNIX Systems Programming, Kay Robbins and Steven Robbins, Prentice Hall, 2003
7. K. L. Calvert and M. J. Donahoo, TCP/IP Sockets in Java: Practical Guide for Programmers (The Practical Guides Series), Morgan Kaufmann Publishers, October 2001
8. TCP/IP and Linux Protocol Implementation, Crowcroft and Phillips, Wiley, 2002.
9. W. Richard Stevens, TCP/IP Illustrated, Volume 1: The Protocols, Addison-Wesley, 1994 (ISBN: 0201633469)

E-resources and support tools

1. CISCO on-line module CISCO Networking Books www.netacad.com
2. Packet Tracer – Network Simulator
3. GNS3 – Network Simulator